

EXPLANATION OF SIGNIFICANT DIFFERENCE

NORTHERNAIRE PLATING COMPANY KYSOR OF CADILLAC, INC. CADILLAC, MICHIGAN

I. Introduction

The Cadillac Industrial Park encompasses approximately three-fourths of a square mile within the city of Cadillac, in Wexford County Michigan. The park is bounded by Thirteenth Street to the north, Mitchell Drive to the east, Wright Street to the south, and Leeson Avenue to the west. Currently, approximately 40 industries operate in the industrial park. There are two Superfund sites in the industrial park: the Northernnaire Plating Company site (Northernnaire), and the Kysor of Cadillac site, (Kysor) (Figure 1).¹

Northernnaire is a former electroplating facility which operated on approximately 13 acres of land located at the corner of Sixth Street and Eighth Avenue within the industrial park.

The Kysor facility is an active manufacturer of truck parts located at the southern end of the industrial park on Wright Street.

The U.S. Environmental Protection Agency (U.S. EPA) and the Michigan Department of Natural Resources (MDNR) are the lead and support agencies, respectively for the remedial design and remedial action at the Northernnaire/Kysor site. The U.S. EPA and the MDNR are acting under the authority of the Comprehensive Environmental Response, Compensation, and Liability Act of 1980, as amended (CERCLA), 42 U.S.C. § 9601, et seq. On September 29, 1989, the U.S. EPA issued a Record of Decision (ROD) for both the Northernnaire/Kysor site. The ROD for Northernnaire outlined the remedy selection process and the selected cleanup action for the contaminated groundwater at the Northernnaire site. The ROD for Kysor outlined the selected remedy for both the soils and the groundwater at the Kysor site.

This Explanation of Significant Difference (ESD) documents the final decision by EPA to include the North Park and West of Leeson areas of contamination in the Northernnaire/Kysor site.

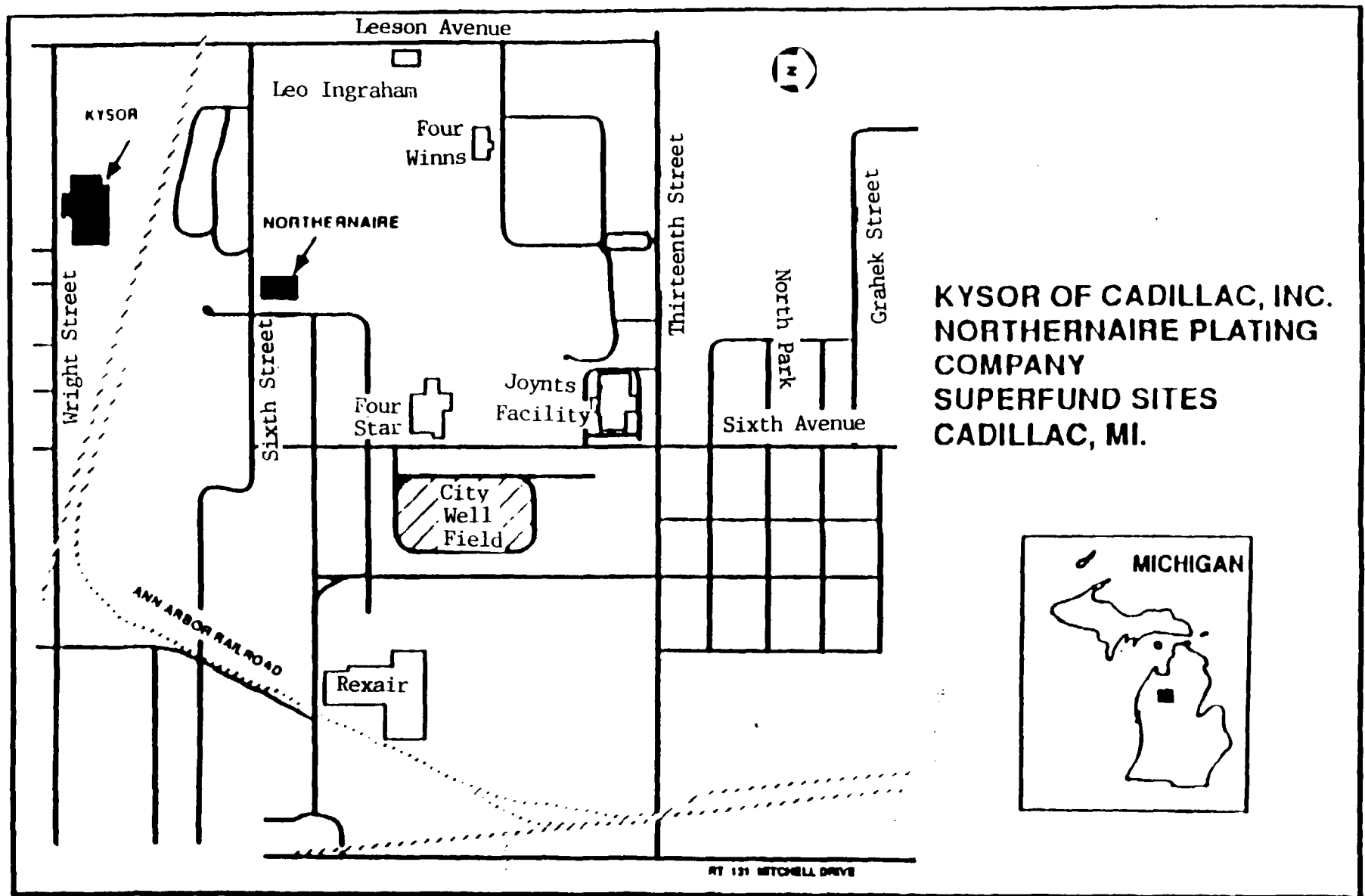
II. Requirement to Address Significant Change

Pursuant to Section 300.435 of the National Contingency Plan, 40

¹ The Northernnaire site and the Kysor site are jointly referred to as "Northernnaire/Kysor site".



Figure 1



CFR § 300.435, U.S. EPA as the lead agency, after appropriate consultation with the State, may determine that a significant change to the selected remedy described in the ROD is warranted.

Section 117(c) of CERCLA, requires that:

After adoption of a final remedial action plan [ROD]:

- (1) if any remedial action is taken,
- (2) if any enforcement action under section 106 is taken, or
- (3) if any settlement or consent decree under section 106 or section 122 is entered into,

and if such action, settlement, or decree differs in any significant respects from the final plan, the [U.S. EPA or the State] shall publish an explanation of the significant differences [ESD] and the reasons such changes were made.
42 U.S.C. §9617(c).

In this case, the U.S. EPA, after appropriate consultation with the State, determined that an ESD was appropriate to explain and document its final position with respect to defining the appropriate extent of the Northernnaire/Kysor site. This ESD document and all of the technical information and data relating to it shall become part of the administrative record for the sites, which is available for viewing at the Cadillac Wexford Library, and at the U.S. EPA regional offices in Chicago, Illinois, during normal business hours.

III. Site History

A. Northernnaire

Northernnaire, located at the northwestern corner of Sixth Street and Eighth Avenue, operated from 1971 to 1981, providing custom chrome and nickel plating finishes to automobile and other parts. Improper waste handling and faulty sewer systems were responsible for releasing contamination including hexavalent chromium, cadmium, and cyanide to the surrounding soils. As a result of leaching through the soils to the groundwater, hexavalent chromium was detected in private wells downgradient of Northernnaire in 1978. The city revoked Northernnaire's discharge permit and ordered them to conduct a hydrogeologic study. In 1979 a plan for the study was submitted but Northernnaire later indicated that no funds were available for the investigation. Northernnaire was purchased by Top Locker Enterprises of Florida. A 1981 site inspection revealed that the facility was apparently out of business.

The site was included on the National Priorities List (NPL) in July 1982 primarily based on concerns for the possible impact of

wastes on the municipal wellfield, and the potential for direct contact threats.

Two children playing in the area of the Northernaire facility received chemical burns from the hazardous wastes stored in containers outside the building. Emergency Response measures were conducted by U.S. EPA and MDNR shortly thereafter in 1983, including removal of waste drums, process liquids and sludges, and a section of the sewer line.

In 1984 Region V referred an action to the Department of Justice for filing a civil complaint to recover all "response costs" associated with the removal action. The District Court for the Western District of Michigan rendered a judgement granting U.S. EPA all of its response costs including prejudgment interest totalling \$342,823.22. One of the defendants, R.W. Meyer appealed that case to the 6th Circuit Court of Appeals. The District Court's decision was affirmed by the Court of Appeals.

A Remedial Action Master Plan for the Northernaire site was completed in 1983. A Remedial Investigation was completed in 1984-1986. At that time, U.S. EPA decided to proceed with a source control operable unit in order to expedite remediation of the soil contamination and continue investigation of the groundwater contamination. A Focused Feasibility Study of proposed source control measures was completed by MDNR in 1985. A Record of Decision (ROD) for the source control operable unit (#1) was signed on September 11, 1985. The ROD included excavation of contaminated soil and sewerline sediments and disposal at an off-site facility, limited modification of the building, and restoration of the sewerline. A Remedial Design for the source control operable unit was completed in 1987 and the remedial action was completed in 1991.

B. Kysor of Cadillac, Inc.

The Kysor facility is a large manufacturer of truck parts located on Wright Street at the southern edge of the industrial park. This facility commonly uses solvent cleaners and degreasers. Past disposal practices included dumping barrels of spent solvents directly onto the soils behind the plant. The site was proposed for the NPL in September of 1985.

Kysor performed hydrogeologic studies in 1980 and 1983. Kysor installed several monitoring wells throughout the industrial park and in the vicinity of the city wellfield. Analysis of groundwater indicated elevated levels of TCA, TCE, PCE, DCE, DCA, xylene, toluene, ethylbenzene, and chloroform.

Partial remediation of contaminated soils was conducted by Kysor in 1981. Approximately 700 cubic yards of contaminated soil were removed. MDNR observers noted strong odors remaining in the

excavation pit and samples indicated that elevated levels of TCE, TCA, PCE, ethylbenzene, and xylenes remained in the soils.

The Remedial Investigation/Feasibility Study (RI/FS) was a state-lead, state-funded project and was completed in September 1989. A ROD for both the Northernnaire and Kysor sites was signed on September 29, 1989. The RI/FS identified several other facilities located within the industrial park as sources of groundwater contamination at the Kysor site. These include: Four Winns, Inc. (Frisbie Street facility), Leo Ingraham, Sr. property, Four Star Corporation, and Joynts/Four Winns.

A Unilateral Order (Order) was issued to the PRPs to perform the remedial design (RD) for the selected remedy at the Northernnaire/Kysor site. Kysor is performing the work required by the Order, and is currently preparing the 95% design document.

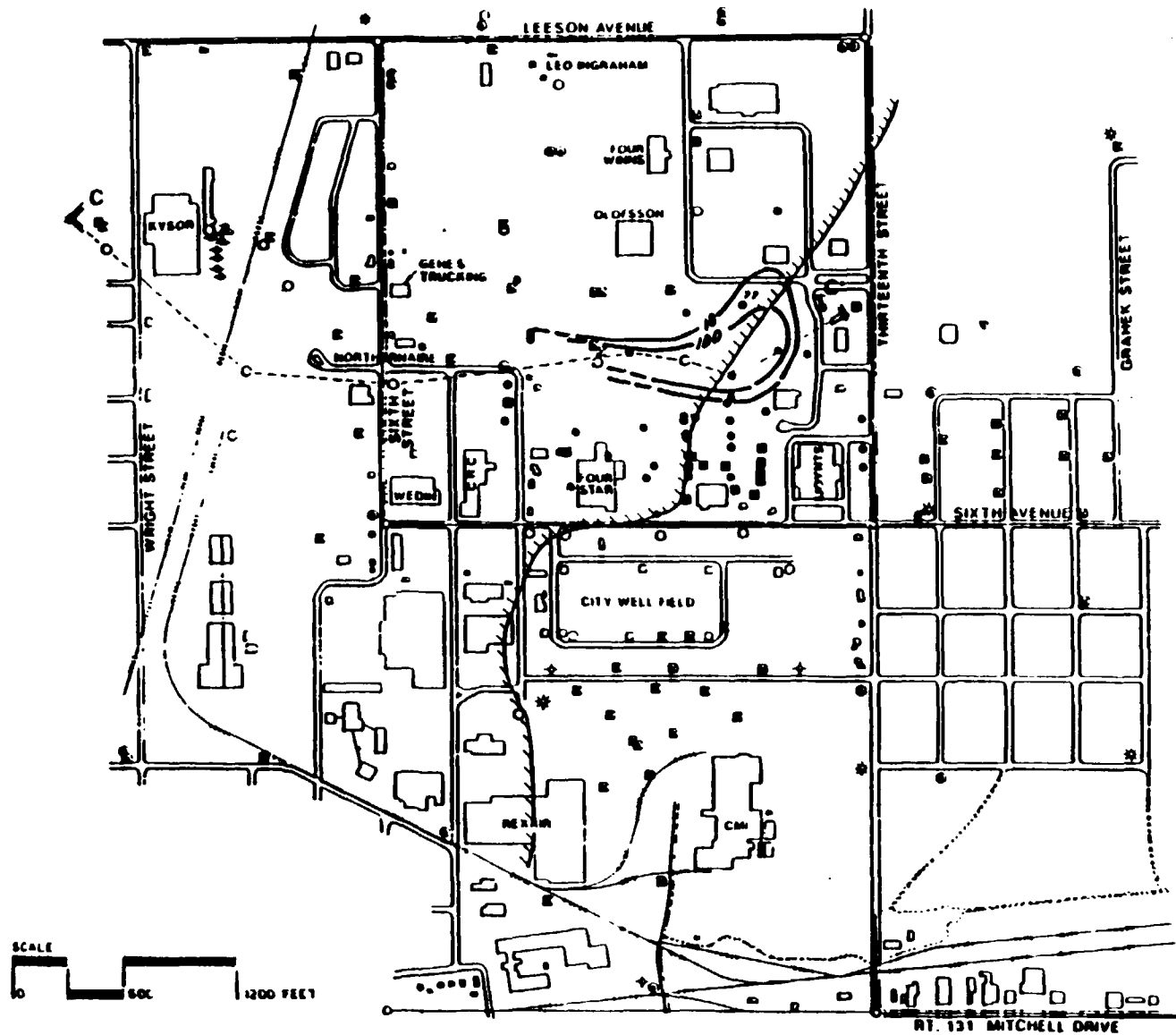
C. Record of Decision

The major components of the remedy listed in the September 29, 1989 ROD for the Northernnaire/Kysor site include:

- Taking appropriate action to ensure that current and future landowners do not use the contaminated groundwater aquifers as a source of drinking water. Activities at the Kysor site will be controlled to prevent new contaminant releases from the site.
- Constructing a fence around the Kysor site to prevent trespassing.
- Construction of a groundwater extraction and treatment system. The VOC contamination in the groundwater shall be treated using an air stripping with vapor-phase carbon adsorption method. The hexavalent chromium in the groundwater shall be treated using a carbon adsorption method.
- Installation of a vapor extraction system to remove the VOC contamination from the soils at the Kysor site.

IV. Significant Difference

The groundwater contamination associated with the Northernnaire/Kysor site was delineated in the Remedial Investigation (RI) for these sites (Figures 2, 3 & 4). The RI identified VOC contamination north of Thirteenth Street in an area referred to as North Park. This area is actually the North Park and Pine View subdivisions; however, for consistency this ESD continues to use the term "North Park" for this area. During the early stages of the design process, and consistent with the ROD, U.S. EPA expected the design and subsequent remedial action

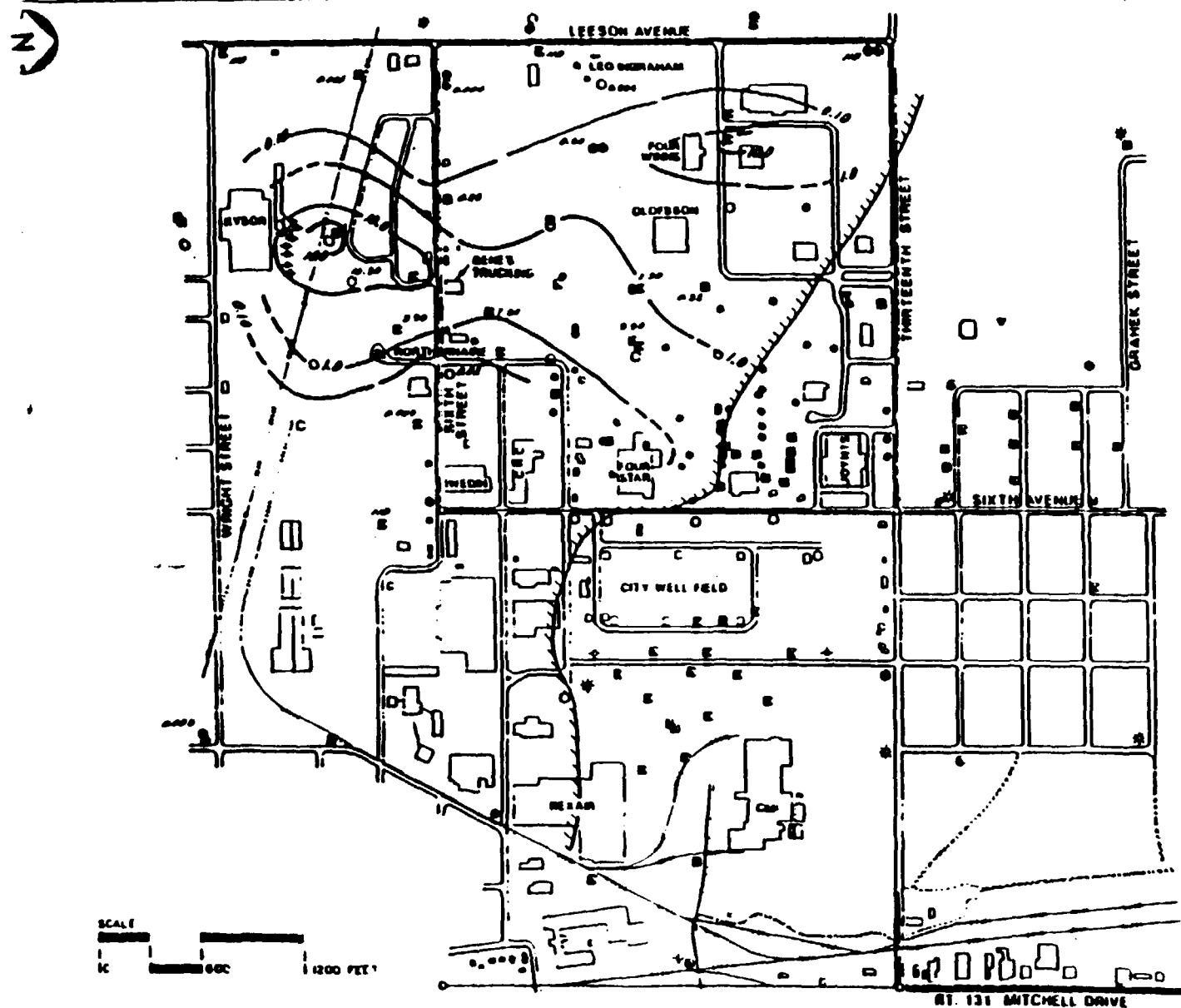


LEGEND

- FOUR STAR WELLS
- DNR WELLS
- KYSOR WELLS
- M.W. WELLS (NORTHERNAIRE)
- CITY WELL HOUSES
- SOURCE LOCATION WELLS
- GENERAL WELLS
- ▼ PIEZOMETERS
- ★ MUD BORINGS
- ✦ KYSOR SOIL BORINGS
- ✦ MUD ROTARY WELLS
- - - - - HEX-CHROME CONCENTRATION CONTOUR IN PPB
- ||||| NORTHERN EXTENT OF UPPER AQUIFARD
- ^{***} HEX-CHROME IN PPB

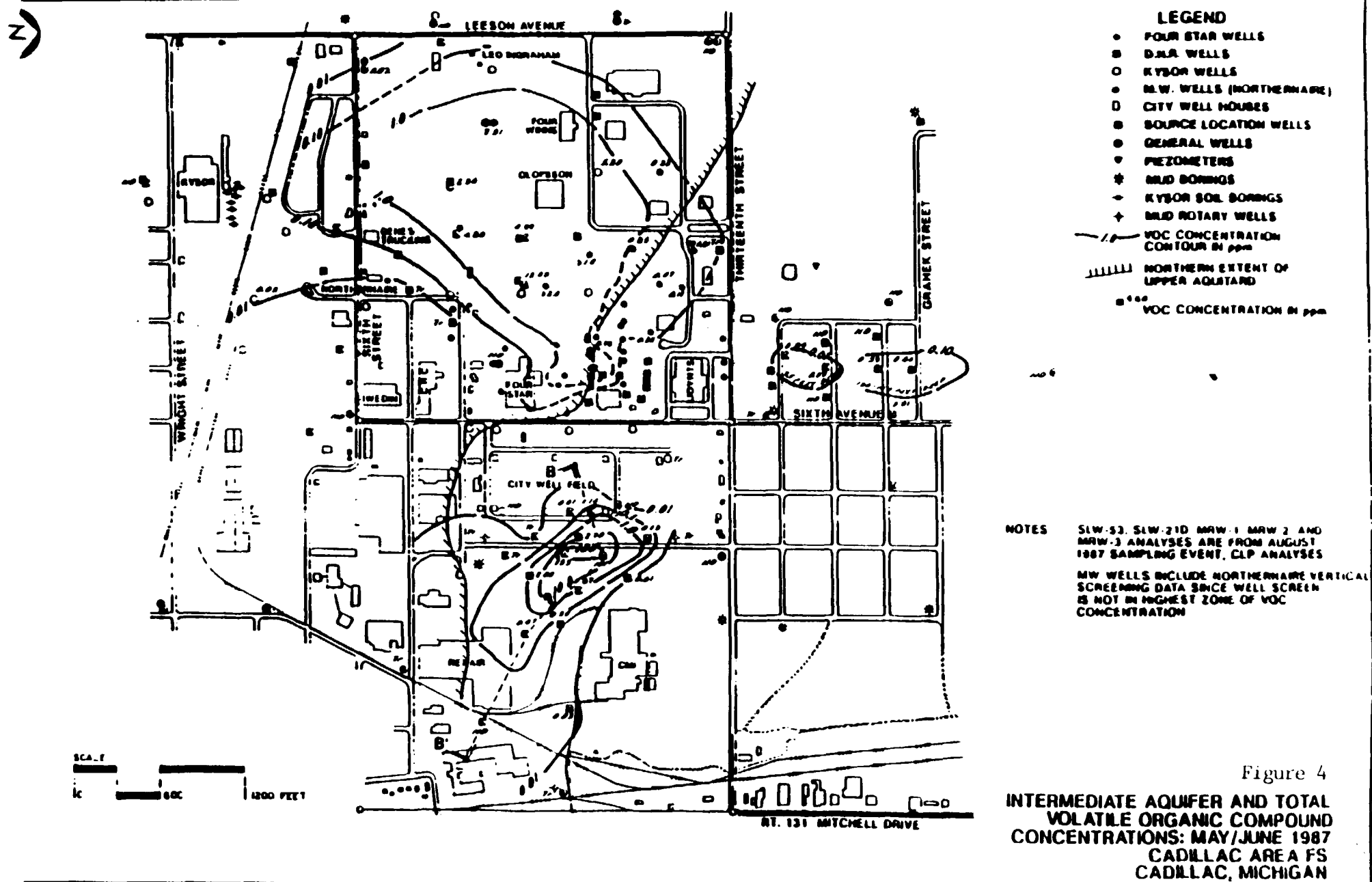
NOTE: CONCENTRATIONS USED TO DRAW INTERPRETIVE CONTOURS ARE SHOWN AT WELL LOCATIONS
DATA ARE FROM CLP ANALYSES

Figure 2
NORTHERNAIRE PLUMES AND
HEX-CHROME CONCENTRATIONS
MAY/JUNE 1987
CADILLAC AREA FS
CADILLAC, MICHIGAN



NOTE: DATA USED FOR PLUME INTERPRETATIONS ARE CLP ANALYSES

Figure 3
SHALLOW AQUIFER AND TOTAL
VOLATILE ORGANIC COMPOUND
CONCENTRATIONS: JUNE 1987
CADILLAC AREA FS
CADILLAC, MICHIGAN



to include the North Park area of contamination.

Upon further review of the available technical data, U.S. EPA was unable to confirm the extent to which contaminated groundwater in the North Park area and the West of Leeson area was commingled with contaminated groundwater at the Northernnaire/Kysor site. The basis for that determination was an apparent "clean zone" suggesting a possible separation between the two contaminated areas. Since the degree of commingling between the areas of contamination was not conclusive, and because additional sampling information was not yet available, the connection between the site and North Park and West of Leeson areas was unclear. Therefore, by letter dated September 4, 1992, U.S. EPA indicated that the Remedial Design need not address the North Park and West of Leeson areas of contamination. In its September 4, 1992 letter, U.S. EPA also stated that if future data indicated that contamination at the site was commingled with contamination at North Park and the West of Leeson areas, the design would be revised to address both areas of contamination.

Based upon sampling by MDNR in September 1992 and MDNR's report dated June 11, 1993 (Appendix A), and after a thorough review of previously generated technical data, U.S. EPA finally concludes that groundwater contamination in the North Park and West of Leeson areas is commingled with the groundwater contamination at the Northernnaire/Kysor site. Therefore, remediation of the site shall include contaminated groundwater in the North Park and West of Leeson areas.

This ESD does not document a change to the ROD remedy with respect to including the North Park and West of Leeson areas in the Northernnaire/Kysor site since the ROD addresses those areas as signed. However, in light of U.S. EPA's letter of September 4, 1992, this ESD documents U.S. EPA's final position that the Northernnaire/Kysor site includes North Park and West of Leeson areas of contamination. Furthermore, due to the migration of contaminated groundwater at the Northernnaire/Kysor site, and to continuing investigation of the contamination plume, other areas of contamination, which are commingled with the Northernnaire/Kysor Site, may also be included in the site and addressed by this remedial action.

V. Affirmation of Statutory Determination

The selected remedy, which addresses the North Park and West of Leeson areas, remains protective of human health and the environment and complies with federal and state requirements that are applicable or relevant and appropriate to this remedial action. In addition, the selected remedy is cost-effective and utilizes permanent solutions and alternative treatment technologies to the maximum extent practicable for this site.

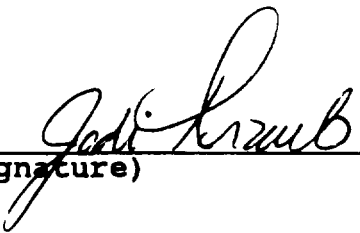
VI. State Comment

The State of Michigan was consulted regarding this change and has reviewed this ESD. The State agrees that the Remedial Action must include the North Park and West of Leeson areas of contamination; however, the State continues to dispute the established clean-up levels for TCE in the ROD.

VII. Public Participation Activities

This ESD and other documents related to this project are available for public review at the Northernnaire/Kysor site information repository located at the Cadillac Wexford Library, and at the U.S. EPA regional offices in Chicago, Illinois, during normal business hours.

APPROVED


(Signature)

3/3/94
(Date)

DISAPPROVED

(Signature)

(Date)

Attachment A

MICHIGAN DEPARTMENT OF NATURAL RESOURCES

INTEROFFICE COMMUNICATION

June 11, 1993

TO: Jeremy Firestone, Assistant Attorney General
Department of Attorney General

FROM: Kevin Kincare, Senior Hydrogeologist
Geological Survey Division

SUBJECT: Fall 1992 Cadillac Area Groundwater Investigation

This memorandum report is a summary of data collected by the Michigan Department of Natural Resources (MDNR) in September 1992 to delineate the limits of contamination in the Cadillac Industrial Park (CIP). The data collected during that sampling effort was previously provided to EPA. The data was collected partly in response to investigations by Kysor that showed significant migration of trichloroethene (TCE) beyond the boundaries established by the 1986/87 MDNR sponsored study. In particular, samples taken from Kysor well F-2 (fig 1), which showed as much as 525 ppb TCE (table 1). In fact, contamination was found to a depth of 229 ft at the top of the city aquifer confining layer. This well is located over 100 ft north of Thirteenth St. in the catholic church parking lot. The northern limit of the plume in this vicinity, in 1987, was 70 ft south of Thirteenth St.

The North Park portion of MDNR's 1992 investigation consisted of sampling 18 existing wells, drilling and sampling 2 new wells, an elevation survey and static water levels measurements. Four existing wells in North Park (north of Thirteenth St.) were still contaminated with TCE (table 2). GW-11 (Seventh Ave. and Fourteenth St.) was non-detect at a depth of 54 ft. A new well, MW-22, was drilled at the GW-11 location. This well was vertically sampled every 10 ft beginning at 43 ft. TCE was first detected at the 83 ft level, reached a high at the 133 ft level, and returned to non-detect at 163 ft. Another well, MW-20, was drilled between MW-22 and F-2. This well was also contaminated with TCE from a depth of 88 ft to the bottom of the boring at 188 ft. A cross section of wells GW-17, DNR-11, SLW-2, MW-22, MW-20, F-2 show a continuous zone of contamination from Sixth Ave. to the west side of the catholic church parking lot, over 1100 ft across.

Jeremy Firestone
June 11, 1993
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Also of interest to the MDNR, was the October 1991 sampling of wells in the vicinity of Four Winns/Joynts. These wells showed the expansion of the Four Star/Kysor plume onto the Four Winns/Joynts property. Some of these wells were resampled for the Fall 1992 study. Sample results (table 2) confirm that previously uncontaminated wells upgradient of Four Winns/Joynts, are now contaminated.

Cogen Michigan, Inc (CMI) also sponsored a site assessment of the proposed power/steam plant in the west Leeson St. area. This investigation also yielded data indicating ground water contamination. In particular, CMS-3, had water samples (table 3) that showed 17 ppb TCE and 34 ppb 1,1,1 trichloroethane (1,1,1 TCA) 500 ft west of Leeson St (fig 1). The west Leeson area was investigated by R.W. Beck Associates in 1991. Beck showed TCE in CMS-3, 500 ft from Leeson Ave, while CMS-6, only 250 ft away, was non-detect for TCE. There was 8 ppb 1,1,1 TCA at 129 ft in CMS-6. To investigate the west Leeson area, the MDNR sampled 11 existing wells, drilled and vertically sampled 2 new wells, surveyed elevations and read static water levels. The MDNR drilled MW-21, 200 ft south of CMS-6 (still 250 ft west of Leeson Ave.). This well had TCE contamination from 178 to 196 ft. MW-23 was then drilled near SLW-16, 50 ft east of Leeson Ave. This well was contaminated with TCE from a depth of 113 ft to the end of the boring at 193 ft. A cross-section of wells GW-10d, MW-23, SLW-16, MW-21, and CMS-3 show an unbroken contaminated profile over 1200 ft long.

cc: Cornelius, MDNR
Croskey, MDNR
Cunningham/Roycraft, MDNR
Zakutansky, EPA

TABLE 1 Analyte to Presume

taken from Northernard/Kysor Sites, Wexford County, Michigan, Remedial Design Additional Studies, July 7, 1991

Kysor Industries
Groundwater - Vertical Profiling
Analytical Results

Volatile Organic Compounds

Sample ID Boring No.:	GPSF2	GPS-F2-FB-1	GPSF2	GPSF2	GPSF2	GPSF2	GPSF2FB2	GPSF2	GPSF2	GPSF2
Sample ID Depth Interval:	70-71	--	79-80	89-90	99-100	109-110	--	118-119	128-129	138-139
Sample Type:	Duplicate	Field Blank-1	Investigative	Investigative	Investigative	Investigative	Field Blank	Investigative	Investigative	Investigative
Sample Date:	04/30/91	04/30/91	04/30/91	04/30/91	05/01/91	05/01/91	05/01/91	05/01/91	05/02/91	05/02/91
Parameter										
chloromethane	ug/l	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U
vinyl chloride	ug/l	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U
bromomethane	ug/l	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U
chloroethane	ug/l	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U
trichlorofluoromethane	ug/l	10 U	10 U	10 U	10 U	10 U	10 U	10 U	10 U	10 U
1,1-dichloroethene	ug/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.9 U
ethylene chloride	ug/l	2.0 U	2.0 U	2.0 U	2.0 U	2.0 U	2.0 U	2.0 U	2.0 U	2.0 U
trans-1,1-dichloroethene	ug/l	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U
1,1-dichloroethane	ug/l	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U
chloroform	ug/l	0.5 U	0.5 U	0.5 U	0.5 U	0.8 U	1.1 UV	0.5 U	0.7 U	0.7 U
1,1,1-trichloroethane	ug/l	0.5 U	0.5 U	0.5 U	0.7 U	1.2 U	1.8 U	0.5 U	7.2 U	5.1 U
carbon tetrachloride	ug/l	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U
1,2-dichloroethane	ug/l	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U
trichloroethene	ug/l	1.1	1.0 U	1.0 U	1.6 U	1.0 U	3.9 U	1.0 U	2.6 U	1.3 U
1,2-dichloropropane	ug/l	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U
bromodichloromethane	ug/l	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U
chloroethylvinylether	ug/l	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U
cis-1,3-dichloropropene	ug/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
1,1,2-trichloroethane	ug/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
tetrachloroethene	ug/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
dibromochloromethane	ug/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
chlorobenzene	ug/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
bromoform	ug/l	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U
1,1,2,2-tetrachloroethane	ug/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	1.3	0.5 U
benzene	ug/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
toluene	ug/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
ethylbenzene	ug/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
p,m-xylenes	ug/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
o-xylene	ug/l	0.5 U	0.7	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
1,3-dichlorobenzene	ug/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
1,4-dichlorobenzene	ug/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
1,2-dichlorobenzene	ug/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U

TABLE 1 con

ysor Industries
groundwater - Vertical Profiling
Analytical Results

olatile Organic Compounds

Sample ID Boring No.:	GPSF2	GPSF2	GPSF2	GPSF3	GPSF3	GPSF3	GPSF3	GPS-F3-FB	GPSF3	GPSF3
Sample ID Depth Interval:	218-219	218-219	228-229	38.8-39.8	51.7-52.7	60.4-61.4	70.5-71.5	--	80.3-81.3	80.3-81.3
Sample Type:	Investigative	Duplicate	Investigative	Investigative	Investigative	Investigative	Investigative	Field Blank	Investigative	Duplicate
Sample Date:	05/07/91	05/07/91	05/07/91	04/24/91	04/24/91	04/24/91	04/24/91	04/24/91	04/24/91	04/24/91
Parameter										
chloromethane	µg/l	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U
vinyl chloride	µg/l	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U
chloroethane	µg/l	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U
chloroethane	µg/l	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U
trichlorofluoromethane	µg/l	10 U	10 U	10 U	10 U	10 U	10 U	10 U	10 U	10 U
1,1-dichloroethene	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
ethylene chloride	µg/l	2.0 U	2.0 U	0.4 JB	2.0 U	2.0 U	2.0 U	2.0 U	2.0 U	2.0 U
trans-1,1-dichloroethene	µg/l	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U
1,1-dichloroethane	µg/l	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U
chloroform	µg/l	0.4 JB	0.4 JB	0.4 JB	0.5 U	0.6	0.5 U	0.5 U	10.0 V	0.5 J
1,1,1-trichloroethane	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
carbon tetrachloride	µg/l	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U
1,2-dichloroethane	µg/l	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U
trichloroethene	µg/l	106	111	260	1.0 U	1.0 U	1.0 U	1.0 U	1.0 J	1.0 J
1,2-dichloropropane	µg/l	1.0 J	0.2 J	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U
bromodichloromethane	µg/l	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	3.5 V	1.0 U
chloroethylvinylether	µg/l	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 J
cis-1,3-dichloropropene	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
1,1,2-trichloroethane	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 J
tetrachloroethene	µg/l	0.5 U	0.5 U	0.1 J	0.5 U	0.5 J	0.5 U	0.5 U	0.5 U	0.5 U
dibromochloromethane	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.8 V	0.5 U
chlorobenzene	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
bromoform	µg/l	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U
1,1,2,2-tetrachloroethane	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 J
benzene	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
toluene	µg/l	0.5 U	0.5 U	0.5 U	0.5 J	0.9 V	1.3 V	0.5 U	0.5 V	0.6 V
ethylbenzene	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
p,m-xylenes	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
o-xylene	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
1,3-dichlorobenzene	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
1,4-dichlorobenzene	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
1,2-dichlorobenzene	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 J

Sysor Industries
Groundwater - Vertical Profiling
Analytical Results

Volatile Organic Compounds

Sample ID Boring No.:		GPSF2	GPSF2	GPS-F2-FB-3	GPSF2	GPSF2	GPSF2	GPS-F2-FB-4	GPSF2	GPSF2	GPSF2
Sample ID Depth Interval:		149-150	149-150	--	158.5-159.5	149-170	178-179	--	188-189	198-199	208-209
Sample Type:		Investigative	Duplicate	Field Blank	Investigative	Investigative	Investigative	Field Blank-4	Investigative	Investigative	Investigative
Sample Date:		05/02/91	05/02/91	05/02/91	05/02/91	05/02/91	05/03/91	05/03/91	05/03/91	05/06/91	05/07/91
Parameter											
chloromethane	µg/l	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U
vinyl chloride	µg/l	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U
bromomethane	µg/l	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U
chloroethane	µg/l	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U	5.0 U
trichlorofluoromethane	µg/l	10 U	10 U	10 U	10 U	10 U	10 U	10 U	10 U	10 U	10 U
1,1-dichloroethene	µg/l	4.1 V	3.9 V	0.5 U	0.5 U	0.7 V	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
ethylene chloride	µg/l	2.0 U	2.0 U	2.0 JB	2.0 U	2.0 JB	2.0 U	2.0 JB	2.0 U	2.0 JB	0.4 JB
trans-1,1-dichloroethene	µg/l	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U
1,1-dichloroethane	µg/l	1.9 V	1.9 V	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U	0.3 U
chloroform	µg/l	1.4 BV	1.4 BV	0.5 JB	0.5 U	0.5 JB	0.5 JB	0.5 JB	0.5 JB	0.4 B	0.8 B
1,1,1-trichloroethane	µg/l	61.7 V	62.4 V	0.5 U	0.5 U	3.7 V	3.7 V	0.5 U	2.5 V	0.1 J	0.5 U
carbon tetrachloride	µg/l	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U
1,2-dichloroethane	µg/l	1.0 J	1.0 J	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	0.2 J
trichloroethene	µg/l	0.6 V	0.6 V	1 J	525 V	16.2 V	25.4 V	1.0 J	16.1 V	42.6	142
1,2-dichloropropane	µg/l	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U
bromodichloromethane	µg/l	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	0.1 J
chloroethylvinylether	µg/l	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U	3.0 U
cis-1,3-dichloropropene	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
1,1,2-trichloroethane	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.2 J
tetrachloroethene	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 J	0.1 J
dibromochloromethane	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
chlorobenzene	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
bromoform	µg/l	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U
1,1,2,2-tetrachloroethane	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
benzene	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
toluene	µg/l	0.5 U	0.5 U	0.5 J	0.5 U	0.5 J	0.5 U	0.5 J	0.5 U	0.5 U	0.5 U
ethylbenzene	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
p,m-xylenes	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 J	0.5 U	0.5 U	0.5 U	0.5 U	2.0
o-xylene	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 JB	0.5 JB	0.5 JB	0.5 U	0.5 U	0.5 U
1,3-dichlorobenzene	µg/l	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
1,4-dichlorobenzene	µg/l	0.5 U	0.5 U	0.5 J	0.5 J	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
1,2-dichlorobenzene	µg/l	0.5 U	0.5 U	0.5 J	0.5 J	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U

Table 2

Fall 1992 Well Sampling Results

Well	Depth	TCE	1,1,1 TCA	1,1 DCE	PCE
CMS-3	178	260.0	0.0	0.0	0.0
CMS-6	178	2.0	0.0	0.0	0.0
DNR-1	38	2.0	0.0	0.0	0.0
DNR-5	65	2500.0	150.0	7.0	0.0
DNR-7	37	0.0	0.0	0.0	0.0
DNR-8	37	0.0	0.0	0.0	0.0
DNR-10S	40	0.0	0.0	0.0	0.0
DNR-10D	68	18.0	0.0	0.0	0.0
DNR-11	46	8.0	0.0	0.0	0.0
DNR14S	46	11.0	0.0	0.0	0.0
DNR14D	78	720.0	38.0	2.0	25.0
F-2	157	200.0	37.0	3.0	0.0
F-3	87	0.0	0.0	0.0	0.0
FS-2-10	40	2.0	0.0	0.0	0.0
FWMW-4	15	0.0	0.0	0.0	0.0
FWMW-10	35	0.0	0.0	0.0	0.0
GW-8	177	0.0	0.0	0.0	0.0
GW-9S	96	4.0	0.0	0.0	0.0
GW-9D	161	9.0	0.0	0.0	0.0
GW-10S	23	48.0	6.0	0.0	3.0
GW-10D	163	6900.0	200.0	0.0	0.0
GW-11	54	0.0	0.0	0.0	0.0
GW-17	173	69.0	0.0	0.0	0.0
K5S	26	4.0	0.0	0.0	0.0
K5-D	148	6.0	0.0	0.0	0.0
K-10	70	0.0	0.0	0.0	0.0
K-11	48	20.0	0.0	0.0	0.0
K-12	146	0.0	2.0	0.0	0.0
K-13	48	0.0	0.0	0.0	0.0
K-22	48	0.0	0.0	0.0	0.0
KIMW-1	35	12.0	0.0	0.0	0.0
KIMW-2	35	19.0	0.0	0.0	0.0
MW-10	150	2700.0	820.0	80.0	0.0
MW-18	149	320.0	0.0	0.0	0.0
SLW-1	38	0.0	0.0	0.0	0.0
SLW-2	54	6.0	0.0	0.0	0.0
SLW-4	54	11.0	0.0	0.0	0.0
SLW-7	55	0.0	0.0	0.0	0.0
SLW-11	44	6.0	7.0	0.0	0.0
SLW-16	84	7.0	0.0	0.0	0.0
SLW-20	143	58.0	0.0	0.0	0.0
MW-20	68	68.0	0.0	0.0	0.0

Table 2 (cont.)

Well	Depth	TCE	1,1,1 TCA	1,1 DCE	PCE
MW-20	108	37.0	0.0	3.0	0.0
MW-20	188	110.0	0.0	0.0	0.0
MW-20	189	56.0	0.0	0.0	0.0
MW-21	178	56.0	0.0	0.0	0.0
MW-21	188	17.0	0.0	0.0	0.0
MW-21	196	9.7	0.0	0.0	0.0
MW-21	179	72.0	0.0	0.0	0.0
MW-22	113	35.0	0.0	0.0	0.0
MW-22	133	110.0	0.0	0.0	0.0
MW-22	183	2.4	0.0	0.0	0.0
MW-22	134	76.0	0.0	0.0	0.0
MW-23	43	0.0	0.0	0.0	0.0
MW-23	113	850.0	0.0	0.0	0.0
MW-23	153	990.0	0.0	0.0	0.0
MW-23	173	1800.0	0.0	0.0	0.0
MW-23	174	1300.0	0.0	0.0	0.0
MW-24	58	0.0	0.0	0.0	0.0
MW-24	148	0.0	0.0	0.0	0.0
MW-24	158	0.0	0.0	0.0	0.0
MW-24	168	0.0	0.0	0.0	0.0
MW-24	160	0.0	0.0	0.0	0.0

Table 2 - History to Firestone memo 6/9/93

Station from Firestone Beck, Phase I and Phase II Environmental Site Assessment of Cadillac Cogeneration Facility Sites, Cadillac, Michigan
 June 1993

TABLE 6.4-1

Summary of Maximum Detected Concentrations of VOCs in Groundwater

Well (ft/bgl)	Benzene	Ethylbenzene	Toluene	Xylenes	TCE	1,1,1-TCA	1,2-DCP
SAB-1 (123-129) (153-158) (178-183)	ND ND ND	ND ND ND	2 2 3	ND 3 1	ND ND 5*	ND ND ND	ND ND ND
MW-1S (14-19) (120-125) (158-163) (178-183)	1 1 ND ND	ND ND ND ND	2 42* 6 8	ND 2 ND 1	ND ND ND ND	ND ND ND ND	ND ND ND ND
MW-1D (8-13) (127-132) (158-163) (177-182) (201-206)	ND 4* 3* 1 ND	ND ND ND ND ND	ND 10 7 4 ND	ND 2 3 2 ND	ND ND ND ND ND	ND ND ND ND ND	ND ND ND ND ND
MW-2 (17-22) (103-108) (128-133) (163-168) (190-195)	ND 1 1 ND ND	ND ND ND ND ND	ND 2 3 1 ND	ND ND 1 ND ND	ND ND ND ND ND	ND ND ND ND ND	ND ND ND ND ND
MW-3 (27-32) (37-42) (97-102) (123-128) (153-158) (163-168) (173-178) (177-182) (183-188)	3* 1 ND 1 1 ND ND 1 1	3 ND ND ND ND ND ND ND ND	7 2 14 2 ND 1 ND 1 ND	8 6 12 1 ND ND ND 2 1	ND ND ND ND ND ND ND ND 17*	4 ND ND ND ND ND ND ND 34	ND ND ND ND ND ND ND ND 31*

Please note - wells designated MW-# in this table are now called CMS-# to
 comply with Cadillac Area Groundwater site nomenclature

Table 3 cont

TABLE 6.4-1 (Deck)
(continued)

Well (ft/bgl)	Benzene	Ethylbenzene	Toluene	Xylenes	TCE	1,1,1-TCA	1,2-DCP
MW-4							
(14-19)	ND	ND	ND	ND	ND	ND	ND
(125-130)	ND	ND	ND	1	ND	ND	NC
(158-163)	ND	ND	2	ND	ND	ND	ND
(178-183)	ND	ND	ND	ND	ND	ND	ND
(198-203)	ND	1	ND	ND	ND	ND	NC
MW-5							
(5-10)	ND	ND	ND	ND	ND	ND	NC
(19-24)	ND	ND	ND	ND	ND	ND	NC
(53-58)	ND	ND	ND	1	ND	ND	NC
(113-118)	ND	1	3	3	ND	ND	NC
(133-138)	ND	ND	ND	ND	ND	ND	ND
(153-158)	ND	ND	ND	ND	ND	ND	NC
(173-178)	ND	ND	ND	ND	ND	ND	NC
MW-6							
(19-24)	ND	ND	ND	ND	ND	ND	NC
(93-98)	ND	ND	ND	ND	ND	ND	ND
(129-134)	ND	ND	ND	ND	ND	8	NC
(139-144)	ND	ND	ND	ND	ND	ND	NC
(150-155)	ND	ND	ND	ND	ND	ND	ND
(159-164)	ND	ND	ND	ND	ND	ND	NC
(169-174)	ND	ND	ND	ND	ND	ND	NC
(179-184)	ND	ND	ND	ND	ND	ND	NC
(189-194)	ND	ND	ND	ND	ND	ND	NC
(199-204)	ND	ND	ND	ND	ND	ND	NC
GW-8							
(177-182)	ND	ND	ND	ND	ND	ND	ND
MDNR Type B							
Clean-up Levels	1	30	40	20	3	200	5**

*Exceeds MDNR Type B Clean-up Level

**Proposed Maximum Contamination Level (MCL)

ND - Less than method detection limit

TCE - Trichloroethene

TCA - Trichloroethane

DCP - Dichloropropane